Table 1. The Solar System forms from a spinning cloud of gas and dust inside a dense stellar cluster containing around 1000 neighboring stars. Timeline and overview of the most important events (as of July 2026).

	Inner Solar System									
	Mercury		*Venus(!)*		*Earth*		*Mars*		*Asteroid belt*	
10^9a	*10^9a*	*event*	*10^9a*	*event*	*10^9a*	*event*	*10^9a*	*event*	*10^9a*	*event*
4,7										
4,6			4,56	Forms	4,54	Forms	4,6	Forms	4,56	*Protoplanetary Disk*[20]
4,5	4,5	Forms[1]			4,5	Impact, Moon[9]	4,5-4,1	***Pre-Noachian Era:***		
4,4	4,5-3,8	***Pre-Tolstojan &***	4,4	Protoplanet collides[5]	4,4-4	***Hadean Eon***		Crust solidified	4,5-4,1	*Primordial Sorting &*
4,3		***Tolstojan Eras:***	4,39	Crashes back[6]				rapidly[16]		*The Grand Tack*[21]
4,2		Solid crust	4,3-0,7	Habitable[7]						
4,1										
4,0							4,1-3,7	***Noachian Period:***	4,1-3,8	*Mass Depletion*[22]
3,9		Heavily battered			4-2,5	***Archean Eon:***		Rivers and lakes[17]		
3,8					3,85	*PRE-GOE*[10]				
3,7	3,8	***Calorian Era*[2]**								
3,6							3,7-3,0	***Hesperian Period:***		
3,5	3,5	Core cools[3]						Core cooling,	3,5-	*Slow Attrition*
3,4	3,5-	***Mansurian & Kuiperian***						loses its global		
3,3		***Eras*[4]**						magnetic field[18]		
3,2										
3,1										
3,0					3	Cyanobacteria[11]				
2,9							3,0-	***Amazonian Period:***		
2,8								Mars dried out[19]		
2,7										
2,6										
2,5										
2,4					2,5-0,54	***Proterozoic Eon:***				
2,3					2,4	*Great Oxidation Event*				
2,2										
2,1										
2,0										
1,9										
1,8										
1,7										
1,6					1,6	Plant-like organisms				
1,5										
1,4										
1,3										
1,2										
1,1										
1,0										
0,9										
0,8					0,8-0,7	First Animals				
0,7						*Snowball Earth*[12]				
0,6			0,7-	Greenhouse effect[8]	0,63	Bilateral symmetry				
0,5					0,54-	***Phanerozoic Eon:***				
0,4						Cambrian Explosion[13]				
0,3					0,252	Pangea forms				
0,2					0,25	*Great Dying*[14]				
0,1										
0,0					0,066	*Cenozoic Era*[15]				

	Outer Solar System													
	Jupiter		*Saturn*		*Uranus*		*Neptune*		*Kuiper belt*		*Hills cloud*		*Oort cloud*	
10^9a	*10^9a*	*event*	*10^9a*	*event*	*10^9a*	*event*	*10^9a*	*event*	*10^9a*	*event*	*10^9a*	*event*	*10^9a*	*event*
4,7														
4,6	4,6	Forms[23], Grand Tack[24]	4,59	Forms	4,59	Forms[31]	4,59	Forms	4,6	Solar nebula	4,6	Begins forming[37]		
4,5	4,5-4,1	Four largest			4,52	Collision, axial tilt	4,5	Migrate outward	4,5	Fail to coalesce[35]	4,5	Stable inner disk[38]	4,5	*Galactic ejection*[40]
4,4		Galilean moons[25]				retrograde rotation[32]			4,4	*Scattered disc*[36]	4,5-	Steady-state buffer[39]	4,5-	Solidifies[41]
4,3														Functions as
4,2														*cosmic net*[42]
4,1														
4,0	4,1-3,8	Giants shift	4,1-3,8	Giants shift	4,1-3,8	Giants shift[33]								
3,9		*Late Heavy*		*Late Heavy*		*Late Heavy*			4	Objects stabilize				
3,8		*Bombardment*[26]		*Bombardment*[27]		*Bombardment*								
3,7														
3,6														
3,5														
3,4														
3,3														
3,2														
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1,3														
1,2														
1,1														
1,0														
0,9														
0,8														
0,7														
0,6					0,6	Inner ring system[34]								
0,5			0,5	Titan[28]										
0,4														
0,3														
0,2			0,2-0,1	Ring system[29]										
0,1				Modern Titan[30]										
0,0														

Note:

[1]) Planetesimals collide and coalesce to form Mercury, a vast magma ocean covers its surface.
[2]) A massive asteroid impact creates the 'Caloris Basin', one of the Solar System's largest impact basins, magma floods the plains.
[3]) Volcanic 'shutoff'; Mercury's core cools and contracts, causing the crust to wrinkle with massive cliffs and sealing off volcanic vents.
[4]) Slow, intermittent space weathering and occasional meteoroid impacts.
[5]) A massive protoplanet collides with Venus, temporarily granting it a moon.
[6]) The moon orbit decays and crashes back into Venus, this catastrophic impact (likely) reverses its rotation, creating its slow, retrograde spin.
[7]) Liquid water oceans and early plate tectonics potentially create a habitable, Earth-like environment for up to 3 billion years.
[8]) Global volcanic activity releases massive amounts of carbon dioxide into the atmosphere. Without active plate tectonics or water to trap carbon, a runaway 'greenhouse effect' takes hold, oceans boil away completely.
[9]) A giant protoplanet collision 'tilted' Earth's axis, created 24-hour days and formed the Moon from the broken debris.
[10]) No O_2 in the atmosphere.
[11]) First to perform photosynthesis, leading to the 'Great Oxygenation Event', completely transforming Earth's atmosphere.
[12]) Drastic climate drops encased the entire planet in ice multiple times, forcing surviving organisms to adapt to extreme conditions.
[13]) A massive burst of evolution occurs, establishing nearly all modern animal phyla, including mollusks, arthropods and chordates.
[14]) Triassic Era, after the 'Great Dying' extinction (the Permian–Triassic extinction event), the first dinosaurs and early mammals evolve.
[15]) The modern Era, a 'mass extinction' (the Cretaceous–Paleogene extinction event K–Pg) wipes out non-avian dinosaurs, allowing mammals to rapidly diversify and become dominant.
[16]) The planet's crust solidified rapidly around 4.54 billion years ago. The planetary 'dichotomy' (the northern lowlands and southern highlands) formed via massive asteroid impacts during this phase.
[17]) The planet possessed a thick atmosphere, a protective magnetic field and extensive surface rivers and lakes.
[18]) Global core cooling caused the planet to lose its global magnetic field, subsurface water releases carved immense outflow channels across the surface.
[19]) Mars dried out completely into an arid desert, dominated by iron oxide dust storms and glacier-like ice movements. Mars transitioned from a warm, dynamic world to the frozen desert today.
[20]) In the region between proto-Mars and proto-Jupiter, billions of small, rocky and icy chunks called planetesimals begin to condense. As Jupiter fully forms, its immense gravitational pull begins to disrupt the region, the 'Jupiter Interruption', instead of gently fusing into a single new planet, the planetesimals collide violently and shatter.
[21]) According to planetary migration models like the 'Grand Tack', Jupiter moves inward toward the Sun and then back out. This cosmic sweeping flings dry, rocky bodies from the inner Solar System outward, and icy, carbon-rich bodies from the outer Solar System inward, mixing them together into the belt today.
[22]) A major orbital shift among the giant planets (the 'Late Heavy Bombardment' or 'Nice Model' period) destabilizes the belt. Gravitational resonances eject over 99% of its original mass, hurling rocks into the inner Solar System and heavily cratering Earth and the Moon.
[23]) Jupiter, the oldest planet, forms a solid core of silicates and ice beyond the solar system's 'snow line' and begins rapidly vacuuming up surrounding hydrogen and helium gas.
[24]) Jupiter's interaction with the solar gas disk drags the planet inward toward the Sun (to where Mars sits today). Saturn's gravitational pull later drags Jupiter back out to its current position.
[25]) Jupiter develops its own disk of debris, which 'condenses' into its four largest Galilean moons: Io, Europa, Ganymede and Callisto.
[26]) Jupiter and the other gas giants shift orbits, this destabilizes the Solar System, triggering a period of chaotic asteroid impacts known as the 'Late Heavy Bombardment'.
[27]) The 'Late Heavy Bombardment' delivers volatile-rich materials, helping to establish Titan's unique organic chemistry.
[28]) Vigorous cryovolcanism (ice volcanoes) dominates Titan, continually resurfacing the moon and erasing early impact craters.
[29]) Recent scientific consensus suggests Saturn's ring system may have formed dynamically during this window from a destroyed icy moon.
[30]) A massive collision between two older moons potentially merges to form modern Titan, inadvertently generating the debris that creates Saturn's iconic rings.
[31]) Uranus forms from the solar nebula as a massive 'ice giant' composed primarily of water, ammonia and methane.
[32]) An Earth-sized planetesimal collides with Uranus, which massive impact permanently knocks the planet onto its side, giving it an extreme 97.8° axial tilt and a retrograde rotation.
[33]) Jupiter and Saturn shift orbits, pushing Uranus and Neptune further out into the frozen edges of the Solar System.
[34]) Gravitational interactions disrupt some inner moons, the debris shatters to form Uranus' narrow, dark inner ring system.
[35]) Icy planetesimals fail to coalesce into a large planet due to low material density at the outer edge.
[36]) Neptune moves outward, scattering up to 99% of the primordial belt's original mass into deep space and creating the 'scattered disc'.
[37]) The Hills cloud begins forming alongside the solar system. Giant planets gravitationally scatter icy planetesimals outward into distant orbits.
[38]) Galactic tides and passing birth-cluster stars circularize these chaotic orbits, which creates a dense, stable inner disk
[39]) It constantly repopulates the easily disrupted outer Oort cloud with trillions of icy bodies.
[40]) The young Sun is ejected from its native birth cluster and begins to 'navigate' the Milky Way galaxy as an isolated, independent planetary system.
[41]) As the Solar System leaves its cluster, the massive gravitational influence of neighboring stars fades. The 'galactic tide', forces exerted by the Milky Way's disk, and passing stars stabilize the thrown-out debris, which shapes the scattered icy bodies into the spherical outer Oort Cloud.
[42]) It captures comet-like objects originally belonging to other stellar systems that pass too close to the Sun.

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